

Spatio-Temporal Evolutionary Graph Learning for Brain Network Analysis Using Medical Imaging

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Abstract—Dynamic functional brain network (DFBN) can flexibly describe the time-varying topological connectivity patterns of the brain, and show great potential in brain disease diagnosis. However, most of the existing DFBN analysis methods focus on capturing the dynamic interaction at the brain region level, ignoring the spatio-temporal topological evolution across time windows. Moreover, they are difficult to suppress interfering connections in DFBNs, which leads to a diminished capacity for discerning the intrinsic structures that are intimately linked to brain disorders. To address these issues, we propose a topological evolution graph learning model to capture disease-related spatio-temporal topological features in DFBNs. Specifically, we first take the hubness of adjacent DFBN as the source domain and the target domain in turn, and then use Wasserstein distance (WD) and Gromov-Wasserstein distance (GWD) to capture the brain's evolution law at the node and edge levels, respectively. Furthermore, we introduce the principle of relevant information to guide the topology evolution graph to learn the structures that are most relevant to brain diseases yet least redundant information between adjacent DFBNs. On this basis, we develop a high-order spatio-temporal model with multi-hop graph convolution to collaboratively extract long-range spatial and temporal dependencies from the topological evolution graph. Extensive experiments show that the proposed method outperforms the current state-of-the-art methods, and can effectively reveal the information evolution mechanism between brain regions across windows.

Index Terms—Dynamic brain network, spatio-temporal evolution, Wasserstein distance, graph convolution, brain disease diagnosis.

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I. INTRODUCTION

THE brain is a complex dynamic system, with continuous fluctuations in information communication between regions even in a resting state [1]. Functional magnetic resonance imaging (fMRI) detects the changes in blood oxygen level-dependent signals, capturing the spontaneous interactions between these brain regions, and it is often used to construct functional brain networks [2]. Compared with the static functional brain network, the DFBN offers a time-varying and comprehensive perspective for studying the brain's cognitive ability and connectivity pattern. Studies indicate that some neurodegenerative diseases can alter the spatio-temporal properties of DFBNs [3]. Therefore, developing effective spatio-temporal topology analysis methods for DFBNs is crucial for brain disease diagnosis and biomarker discovery.

To effectively describe the evolution process of brain networks, various DFBN construction methods have been proposed. For instance, Wang et al. segmented the time series of brain regions into multiple intervals, and then employed the Pearson correlation to estimate the connectivity of each interval [4]. Chen et al. used transfer entropy to measure the temporal information flow between brain regions and constructed DFBNs with causal association [5]. The human brain comprises multiple regions, which may exhibit different activity patterns and durations even when activated simultaneously [6]. Thus, capturing spatio-temporal interactions is essential for understanding the cognitive functions of the brain. For example, Zhu et al. adopted the spatio-temporal variability as the connection weight between brain regions across time domains, and designed a stacked topology preserving DFBNs [7]. Tang et al. segmented fMRI signals into multiple sub-segments, and employed a transformer to learn asynchronous functional interactions [8]. However, these methods focus on analyzing the evolution of DFBNs based on node signal alterations, ignoring the fact that the brain's edge structure also evolves over time. The connection edges, acting as conduits for information transport, not only reflect the neural network's plasticity but also signal transitions in cognitive function and pathological states [9]. Consequently, simultaneously capturing the evolution at both the node and edge levels is crucial for unraveling the mechanisms of brain diseases and improving diagnostic accuracy.

Complex spatio-temporal dependencies support the brain's high-level cognition function, and some neurodegenerative

diseases are often accompanied by symptoms of cognitive impairment. Extracting the intrinsic spatio-temporal correlations within DFBNs is beneficial to the study of pathological states and cognitive transfer. Spatio-temporal graph convolutional networks (GCN) have shown great potential in brain disease diagnosis, as they can cooperatively extract the temporal and spatial features. For example, Dahan et al. leveraged GCN and temporal convolutions to extract the dynamic topology, and achieved promising prediction performance [10]. Liu et al. combined GCN and long short-term memory networks to model the co-occurrence dependence between brain regions across time periods, and attained superior classification results in autism diagnosis [11]. Most of these spatio-temporal models follow the existing message-passing paradigm, which aggregates neighboring node information along the brain graph structure. They took the original DFBN as the path for information interaction, assuming that the existing graph structure can accurately describe the communication pattern between brain regions. However, DFBNs contain a lot of redundant and interfering information [6], [12]. First, the reconstruction of brain networks is a progressive process, and a substantial amount of similar connectivity patterns persist even across different windows, which can lead to information redundancy in brain disease diagnosis [13]. Second, the complexity of brain networks allows the same information to propagate through multiple connections, which may blur the distinction between functional connections that underpin normal cognitive processes and abnormal connections caused by brain diseases [14], [15]. This interference information can distort the original DFBN from its optimal structure. Therefore, revealing the intrinsic structures closely related to brain diseases in DFBNs is important for effectively aggregating spatio-temporal features.

To address the aforementioned challenges, we develop a topology evolution graph (TEG) learning model to capture the inherent spatio-temporal features in DFBNs. We regard the hubness of adjacent dynamic brain networks as the original distribution and the target distribution in turn, and employ two optimal transport distances: Wasserstein distance (WD) [16] and Gromov-Wasserstein distance (GWD) [17] to simulate the topology evolution process at the node level and edge level, respectively. Here, the optimal transport is defined as the transformation of the brain network from one time window to the next with the most economical cost, and the resulting optimal transport matrix can explicitly capture the evolution across adjacent networks. Specifically, WD measures the distance between node embeddings, while GWD evaluates the graph structure difference by comparing the distance between pairs of nodes within each brain network. By fusing these two distance measures, our method accounts for the evolutionary features of both the node and edge levels, and comprehensively captures the time-varying information of the brain. Furthermore, we introduce the principle of relevant information (PRI) [18] to view the learning of TEG as a trade-off between redundancy reduction and important structure preservation, thus revealing evolution structure related to brain diseases. Finally, we design a high-order spatio-temporal model that combines multi-hop GCN and convolutional neural networks

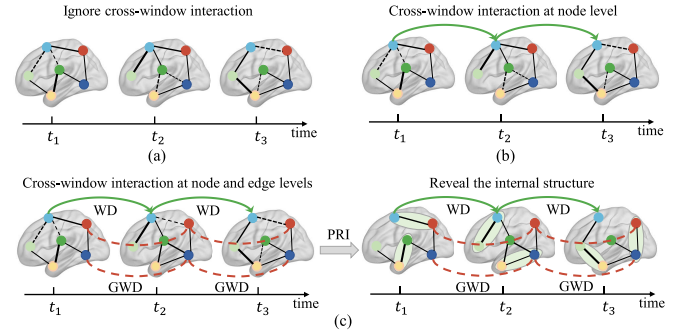


Fig. 1. Comparison of different DFBN analysis strategies. (a) independently analyze each DFBN. (b) cross-window interaction at the node level. (c) The proposed topological evolution graph model can collaboratively capture the cross-window interactions at node and edge levels, and introduce the PRI to learn the intrinsic structure of DFBNs.

to extract long-range spatial dependencies and dynamic temporal features from TEGs.

For comparison purposes, we delineate the distinctions between existing DFBN analysis methods and the proposed method in Figure 1. Specifically, Figure 1(a) shows the methods for analyzing each DFBN independently, which ignores the spatio-temporal interactions that occur within the brain. Figure 1(b) illustrates the DFBN analysis methods that focus on the node level, examining the evolution of brain networks through regional signal changes, yet ignoring the temporal evolution of the edge structure. Different from these methods, as depicted in Figure 1(c), the proposed method synthetically simulates the temporal evolution process at both the node and edge levels. Furthermore, it effectively reveals the intrinsic structure in DFBNs based on the principle of relevant information, which further enhances the aggregation of spatio-temporal features. Briefly, the proposed method has the following contributions:

- (1) In this paper, we propose a topological evolution graph learning model, which not only captures the abundant spatio-temporal topological features in DFBNs, but also effectively reveals the intrinsic structures closely related to brain diseases.
- (2) The proposed method effectively characterizes the dynamic functional topology patterns of brain networks. Specifically, we adopt WD and GWD to capture the evolution law of brain network at the node and edge levels, respectively.
- (3) The topology evolution graph network is developed to further reveal the brain disease related spatio-temporal features. It not only extracts the spatio-temporal discriminative features in the brain network, but also eliminates the redundancy of graph structure.
- (4) Extensive experiments on the public ADNI dataset and the collected Parkinson's disease dataset show that our method has superior classification performance and can provide spatio-temporal biomarkers for brain disease diagnosis.

The remainder of the paper is organized as follows: In Section I, we describe the related work. We then present the proposed method in Section III. Next, we introduce the datasets used and extensive experimental results in Section IV.

In Section V, we discuss the experimental results. Finally, we summarize this work and draw a conclusion.

II. RELATED WORK

A. Optimal Transport

Optimal transport is a mathematical framework for studying probabilistic transformation, which has gained extensive attention in machine learning, such as cross-domain alignment [19], graph neural networks [20], and computer vision [21]. The goal of optimal transport is to transform the initial distribution into the target distribution with the lowest transport cost, and it can effectively measure the dissimilarity between two probabilities. Brain networks can be represented as probability distributions through graph theory or network embedding. Therefore, optimal transport has become a powerful tool for brain network analysis. For example, Ma et al. introduced a sliced Wasserstein graph kernel to evaluate the similarity of functional brain networks, and significantly improved the diagnosability of brain diseases [22]. Liu et al. developed a low-rank sparse representation method with adaptive weights and WD constraints to construct functional brain networks, utilizing the WD between brain regions as the information transport cost to dynamically adjust the sparsity penalty on functional connectivity [23]. Zhu et al. adopted optimal transport to construct high-order dynamic brain networks integrating node evolution information and provided high-order biomarkers for brain disease diagnosis [24].

However, these methods focus on analyzing the functional interactions of brain networks from the perspective of brain region signals, ignoring the temporal and structural information encoded in the topological edge. As a communication bridge between brain regions, edges are the structural basis of brain networks, and they actively participate in cognitive cooperation and information integration in the brain. Therefore, we propose to collaboratively capture the dynamic evolution process at both the node and edge levels by using WD and GWD between pairwise adjacent time windows, which enables us to explicitly observe the information flow in brain networks and helps to reveal the pathological mechanism of brain diseases.

B. Spatio-Temporal Features of DFBNs

The brain network is a dynamic system that constantly reorganizes, and the evolutionary trajectories of its functional connectivity can map the impact of cognitive functions and neurodegenerative diseases on brain connectivity. For instance, Li et al. proposed a cognition-driven topological feature learning strategy that adaptively identifies spatial topological patterns related to cognition within DFBNs [25], [26]. Subsequently, they introduced a multi-scale fusion framework that automatically learns the optimal manifolds of local and global activation patterns, achieving an efficient representation of emotional signals [27], [28]. To further integrate the temporal and topological features in DFBNs, researchers design some spatio-temporal graph models to capture the dynamic dependencies of DFBNs. Spatio-temporal graph models generally use GCN to aggregate the structural features of adjacent nodes,

and then use the temporal convolutional network (TCN) to extract the temporal information flow of the brain:

$$H = TCN(\sigma(GCN(A, X))) \quad (1)$$

where H denotes the spatio-temporal embedding, σ represents the activation function, A and X are the adjacency matrix and node features, respectively. For example, Xu et al. first used three-channel GCN to extract the structural and functional information of DFBN, and then adopted a long short-term memory network to capture the temporal interactions between brain regions [29]. Kong et al. suggested extracting spatial and temporal information from DFBNs by graph attention and temporal fusion network, respectively [30]. Kim et al. initially employed the graph isomorphism network to aggregate neighboring node features, and then used attention mechanisms to extract the spatio-temporal information of DFBNs [31].

However, existing spatio-temporal graph models use the original DFBNs as a path for feature aggregation. DFBNs contain a lot of redundant information, which is different from the real network structure. Therefore, to effectively reveal the intrinsic structure of the brain, we introduce the PRI to guide the generation of topology evolution graphs. In addition, we develop a high-order spatio-temporal GCN to extract the rich spatio-temporal information from topology evolution graphs.

III. PROPOSED METHOD

The framework of the proposed dynamic brain network topology analysis method is presented in Figure 2, which comprises two major components: topology evolution graph learning model and high-order spatio-temporal graph convolutional network. In this section, we present the proposed framework in detail.

A. Topology Evolution Graph Learning Model

As illustrated in Figure 3, our proposed topological evolution graph learning adopts two optimal transport distances: Wasserstein distance for node-level information transfer and Gromov-Wasserstein distance for topological edge transfer. We simulate the brain's evolution process of two levels by transmitting hubness between adjacent networks.

1) *Node Hubness Measure*: In this paper, we assume that the fMRI signal sequence of each subject is $X \in R^{N \times M}$, where N and M denote the number of brain regions and the number of collected time points, respectively. To characterize the dynamic alterations in brain connectivity, we segment fMRI signal sequences using T non-overlapping sliding windows of length $S = M/T$. For the signal subsequence $X_t \in R^{N \times S}$, we use Pearson's correlation coefficient to construct the functional brain network A_t . There are some important nodes in the brain network, which can high-throughput transmit information to other brain regions. To capture the importance of brain regions, we measure the node hubness using PageRank [32] based on the graph topology of the brain network A_t :

$$h_t(u) = \sum_{v \in B(u)} \frac{h_t(v)}{D_v} \quad (2)$$

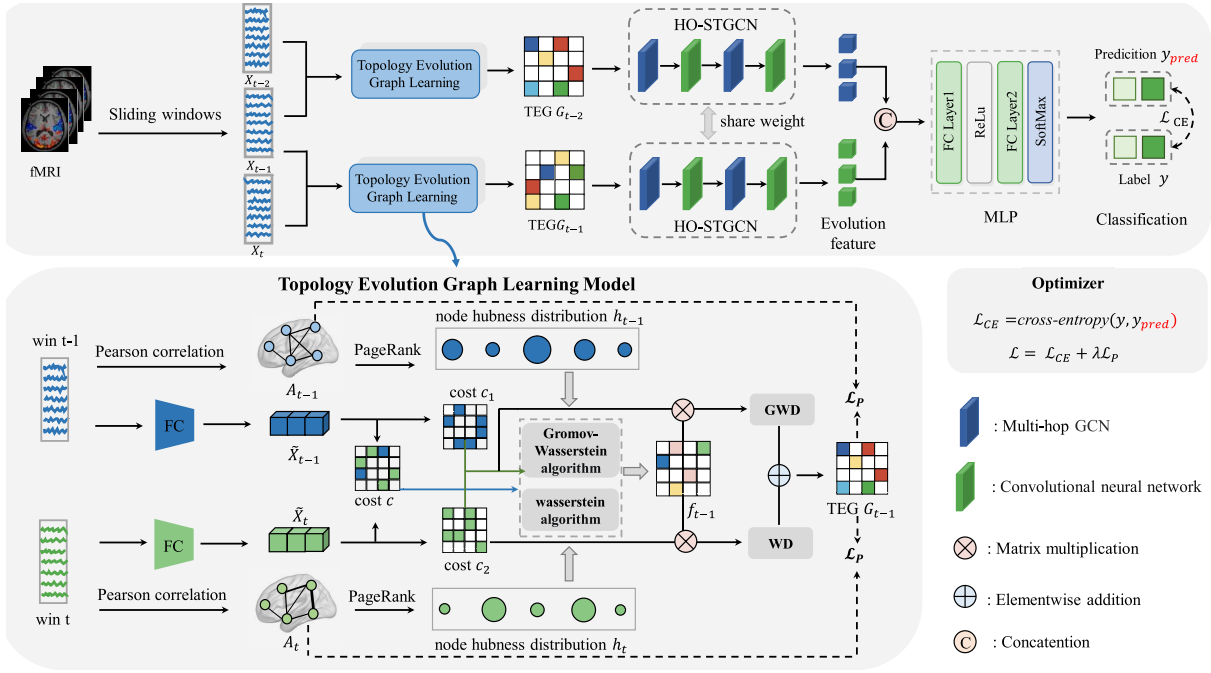


Fig. 2. The framework of the proposed method for brain disease diagnosis.

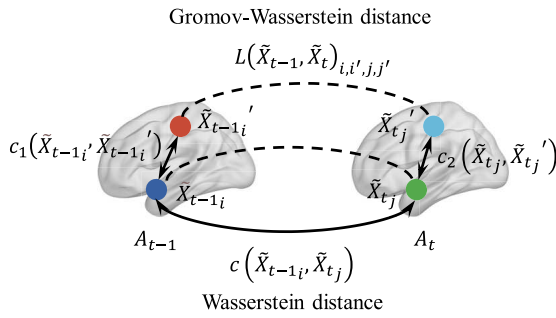


Fig. 3. A detailed illustration of topological evolution graph learning in adjacent brain networks. WD and GWD are used to simulate information transport across windows at the node and edge levels, respectively.

where $h_t(u)$ denotes the hubness of node u in A_t , $B(u)$ is the set of the nodes connected to node u , and D_v represents the degree of node v .

2) *Node-Level Information Transport Through WD*: To capture the evolutionary patterns of brain region activity, we adopt WD to construct hubness propagation models across adjacent brain networks. Specifically, we take the hubness distribution h_{t-1} of the $(t-1)^{th}$ functional brain network as the initial distribution, the hubness distribution h_t of the t^{th} functional brain network as the target distribution, and use WD to find the propagation process of hubness transport from the initial distribution to the target distribution under the most economical case. To adaptively encode the temporal activity information of brain regions, we first use a fully connected layer to encode X_t as $\tilde{X}_t$. Then, we consider the similarity between the node's subsequence of adjacent windows as the transport cost $c(\tilde{X}_{t-1,i}, \tilde{X}_{t,j}) = 1 - \frac{\tilde{X}_{t-1,i}^T \tilde{X}_{t,j}}{\|\tilde{X}_{t-1,i}\|_2 \|\tilde{X}_{t,j}\|_2}$. Therefore, the node-level information transport process f_{t-1} between adjacent

networks can be learned by:

$$\begin{aligned} \mathcal{D}_w(h_t, h_{t-1}) &= \min_{f_{t-1} \in \Pi(h_t, h_{t-1})} \sum_{i=1}^N \sum_{j=1}^N f_{t-1,ij} \cdot c(\tilde{X}_{t-1,i}, \tilde{X}_{t,j}) \\ &= \min_{f_{t-1} \in \Pi(h_t, h_{t-1})} \langle c(\tilde{X}_{t-1}, \tilde{X}_t), f_{t-1} \rangle \end{aligned} \quad (3)$$

where $\Pi(h_t, h_{t-1}) = \{f_{t-1} | f_{t-1} \mathbf{1}_N = h_{t-1}, f_{t-1}^T \mathbf{1}_N = h_t\}$, $\mathbf{1}_N$ is an N -dimensional all-one vector, and $\langle \cdot, \cdot \rangle$ denotes the inner product for matrices.

3) Edge-Level Information Transport Through GWD:

Unlike WD, which directly calculates the information transport between nodes, GWD can be used to calculate the distances between pairs of nodes within each domain, as well as to measure distances between pairs of the same nodes across different domains. Therefore, we further introduce GWD that can fuse edge differences to adaptively capture the topological evolution process of brain networks. Specifically, we consider the structural connectivity differences between adjacent networks as the transport cost of GWD, with the help of the hubness propagation process to capture the complex evolution of brain connectivity. The Edge-level information transport process f_{t-1} between adjacent networks can be learned by:

$$\begin{aligned} \mathcal{D}_{gw}(h_t, h_{t-1}) &= \min_{f_{t-1} \in \Pi(h_t, h_{t-1})} \sum_{i,i',j,j'} f_{t-1,ij} f_{t-1,i'j'} L_{i,i',j,j'} \\ &= \min_{f_{t-1} \in \Pi(h_t, h_{t-1})} \langle L \otimes f_{t-1}, f_{t-1} \rangle \end{aligned} \quad (4)$$

where $L_{i,i',j,j'} = \|c_1(\tilde{X}_{t-1,i}, \tilde{X}_{t-1,i'}) - c_2(\tilde{X}_{t,j}, \tilde{X}_{t,j'})\|_1$ represents the connectivity difference of adjacent networks, where c_i , $i \in [1, 2]$ are functions that evaluate node similarity within the same functional brain network, and $\otimes$ is the matrix multiplication.

In order to effectively integrate the topology evolution law of DFBNs at the node level and the edge level, we promote WD and GWD to share the optimal transport plan f_{t-1} . Therefore, the proposed TEG learning model is as follows:

$$D(h_t, h_{t-1}) = \min_{f_{t-1} \in \Pi(h_t, h_{t-1})} \left\{ \alpha c(\tilde{X}_{t-1}, \tilde{X}_t) + (1 - \alpha)(L \otimes f_{t-1}, f_{t-1}) \right\} \quad (5)$$

where α is used to control the importance of different cost function. We use Sinkhorn algorithm [19] to solve the above optimization problem, to obtain the optimal transport plan f_{t-1} . To comprehensively capture the evolution patterns of adjacent brain networks at the node and edge levels, we use the Wasserstein distance between adjacent brain networks as G_{t-1} :

$$G_{t-1} = \langle \alpha c(\tilde{X}_{t-1}, \tilde{X}_t) + (1 - \alpha)(L \otimes f_{t-1}, f_{t-1}) \rangle \quad (6)$$

4) *Most Relevant Graph Structure Learning With PRI*: In order to learn the evolutionary structure most relevant to brain diseases and alleviate redundant information in TEG, we adopt two adjacent brain networks A_{t-1} and A_t to guide the learning of G_{t-1} with PRI. Specifically, the PRI loss for graph structure learning aims to learn a refined graph G_{t-1} :

$$\mathcal{L}_P = H(G_{t-1}) + \beta(D(G_{t-1} \parallel A_{t-1}) + D(G_{t-1} \parallel A_t)) \quad (7)$$

where $H(G_{t-1})$ measures the redundancy of G_{t-1} , and $D(G_{t-1} \parallel A_i) \{i = t-1, t\}$ denote the distortion level of the learned refined graph G_{t-1} with respect to the network A_i . β is the trade-off between the redundancy reduction of G_{t-1} and its discriminative description power of A_{t-1} and A_t . As β becomes larger, the emphasis is laid more on the distortion term, and more information from A_{t-1} and A_t is preserved in G_{t-1} .

von Neumann entropy [18] regards the graph as a quantum system to quantify the complexity of the graph structure, which can measure the structure redundancy contained in G_{t-1} . For computational convenience, we first add G_{t-1} to its transpose matrix and set the threshold γ to convert its binarization into a symmetric adjacency matrix. For the graph G_{t-1} , it's von Neumann entropy can be defined as:

$$H_v(G_{t-1}) = H_v(\rho) = - \sum_{i=1}^N (\lambda_i \log \lambda_i) \quad (8)$$

where ρ denotes the brain graph density matrix of G_{t-1} , $\{\lambda_i\}$ are the eigenvalues of ρ . The quantum Jensen-Shannon (QJS) divergence [18] measures the distribution difference of quantum states in graph data, we adopt the QJS divergence between the graph density matrices (ρ and ρ_i) of G_{t-1} and $A_i \{i = t-1, t\}$ to evaluate the distortion between them:

$$D_Q(G_{t-1} \parallel A_i) = H_v\left(\frac{\rho_i + \rho}{2}\right) - \frac{1}{2}H_v(\rho) - \frac{1}{2}H_v(\rho_i) \quad (9)$$

where $\rho = \frac{L}{2|E|}$, $L = D_{t-1} - G_{t-1}$ is the Laplacian matrix of G_{t-1} , D_{t-1} denotes the degree matrix of G_{t-1} , and $|E|$ represents the number of edges in G_{t-1} . Similarly, $\rho_i = \frac{L_i}{2|E_i|}$, where $L_i = D_i - A_i$ is the Laplacian matrix of A_i , D_i denotes the degree matrix of A_i , and $|E_i|$ is the number of edges in A_i . Combining the

redundancy term in Eq. (8) and the distortion term in Eq. (9), we can obtain the following objective function:

$$\begin{aligned} \mathcal{L}_P &= H_v(G_{t-1}) + \beta(D_Q(G_{t-1} \parallel A_{t-1}) + D_Q(G_{t-1} \parallel A_t)) \\ &= (1 - \beta)H_v(\rho) + \beta\left(H_v\left(\frac{\rho_{t-1} + \rho}{2}\right) + H_v\left(\frac{\rho_t + \rho}{2}\right) - \frac{1}{2}H_v(\rho_{t-1}) - \frac{1}{2}H_v(\rho_t)\right) \\ &\equiv (1 - \beta)H_v(\rho) + \beta\left(H_v\left(\frac{\rho_{t-1} + \rho}{2}\right) + H_v\left(\frac{\rho_t + \rho}{2}\right)\right) \end{aligned} \quad (10)$$

We ignore $\frac{1}{2}H_v(\rho_{t-1})$ and $\frac{1}{2}H_v(\rho_t)$ in the second line of Eq. (10), because they are a constant in the optimization process.

B. High-Order Spatio-Temporal Graph Convolution

TEG encodes rich topological correlations and temporal dependencies, and effectively extracting the spatio-temporal correlation from TEGs is of great significance for brain disease diagnosis. Traditional spatio-temporal GCN captures the complex topological structures of brain networks by increasing the depth of the network, which consumes a lot of resources and leads to an over-smoothing phenomenon. Therefore, we introduce the multi-hop GCN [33] to aggregate node features at different ranges with fewer parameters. Considering that TEGs can comprehensively represent both temporal and spatial features, we choose 2D convolutional neural network, which is proficient at capturing contextual information, to extract the temporal structural features. The HO-STGCN is defined as:

$$X_{t-1}^l = ReLU\left(W_c * \left(ReLU\left(W_g X_{t-1}^{l-1} \sum_{k=0}^K G_{t-1}^k\right)\right)\right) \quad (11)$$

where W_c and W_g are the learnable weights in CNN and multi-hop GCN, respectively, $*$ represents the 2D convolution operation, K is the order of the neighboring nodes, G_{t-1}^k denotes the k^{th} power of G_{t-1} , and l is the l^{th} layer HO-STGCN. To improve model training efficiency, we employ multiple parameter-shared HO-STGCNs to extract spatio-temporal features from each graph $\{G_{t-1}\}_{t=2}^T$ in the TEGs. In HO-STGCN, we first aggregate spatial features within G_{t-1} using multi-hop graph convolution, and then feed these spatial features into a 2D CNN to extract temporal features.

Finally, we concatenate the spatio-temporal features learned under different windows to obtain the comprehensive embedding, and send it to the multilayer perceptron to predict the diagnosis result. The overall training objective is $\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_P$, where $\mathcal{L}_{CE}$ is the cross-entropy loss for prediction, and λ is the hyperparameters that determine the relative importance of the diseases classification loss and graph learning loss items.

IV. EXPERIMENT

A. Datasets and Preprocessing

1) *Dataset Description*: In this work, we conduct extensive experiments on the publicly available ADNI dataset (<https://adni.loni.usc.edu/>) and the PD dataset collected by the

Affiliated Brain Hospital of Nanjing Medical University. The detailed description of these datasets is as follows. ADNI datasets: We acquired a total of 203 fMRI data for ADNI2 and ADNI3, including 73 healthy controls (HC), 70 patients with significant memory concerns (SMC), and 60 patients with early mild cognitive impairment (EMCI). There are no overlapping subjects between ADNI2 and ADNI3. PD dataset: This dataset collected fMRI data from a total of 162 subjects, consisting of 54 HCs, 44 patients with Tremor-dominant Parkinson’s disease (TDPD), and 64 with Parkinson’s disease with postural instability and gait disorder (PIGD-PD). The fMRI data was collected using a Siemens MAGNETOM Verio 3.0T MRI scanner with the following parameters: repetition time = 2000 ms, echo time = 25 ms, and flip angle $^{\circ}$.

2) *Data Preprocessing*: We perform the same pre-processing operations on two fMRI datasets utilizing the DPARSF toolbox [34]. Specifically, to correct and realign the raw image, we split the serialized data into some pieces and then adjusted them to the EPI template. Next, we employed the detrending technique to reduce the interference of head movements and the influence of cerebrospinal fluid and white matter. Finally, we used the Automated Anatomical Labeling (AAL) atlas [35] to partition all fMRI data in the ADNI dataset into 90 brain regions with 197 time signal points. Considering that the cerebellum plays an important role in the diagnosis of PD, we adopted the AAL atlas to partition all fMRI data in the PD dataset into 116 brain regions with 220 time signal points.

B. Comparison Methods

To verify the performance of our method, we select 12 comparison methods for disease diagnosis, categorized into two groups: based on static brain networks and based on dynamic brain networks. Three methods based on static brain network are included: GCN [36], functional brain network generation (FBNetGen) [37], and complementary graph representation learning for functional network (CGRL) [38]. nine methods based on DFBNs are included: modularity-constrained dynamic representation learning (MDRL) [39], dynamic high-order functional connectivity (DHOFC) [40], high-order dynamic functional connectivity networks (Ho-D-FCNs) [41], spatio-temporal graph neural networks with dynamic functional and effective connection (FE-STGNN) [5], spatio-temporal graph convolution network (ST-GCN) [10], asynchronous common and individual functional brain network (ACIFBN) [8], graph hubness propagation model with optimal transport (GHP-OT) [24], multi-scale spatial-temporal attention networks (MSSTAN) [42], and spatio-temporal interactive graph representation framework (STIGR) [43]. These methods provide a robust benchmark for evaluating the effectiveness of the proposed framework in brain disease diagnosis.

C. Experimental Settings

The proposed method is implemented based on Pytorch using an NVIDIA GeForce RTX3080 GPU. We employ the Adam optimizer with a learning rate set to $4.5e-3$ and the

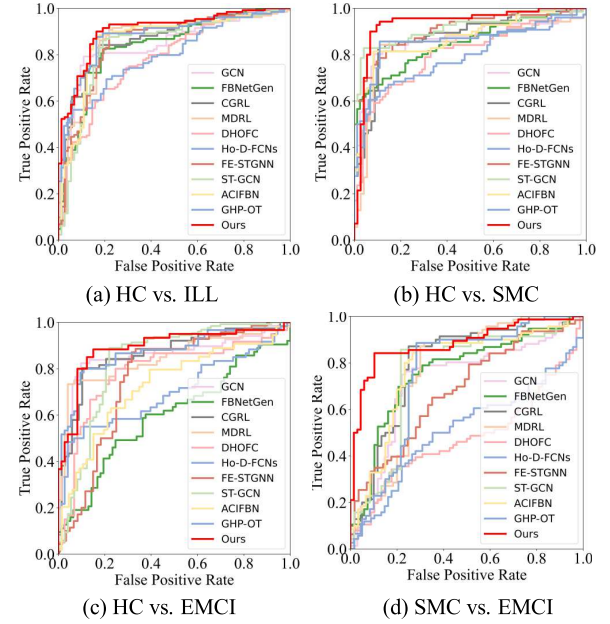


Fig. 4. ROC curves of all methods under different classification tasks.

weight decay is $1e-3$. To alleviate overfitting, we introduce a gradient clipping strategy. In the experiments, we introduce grid search to determine the model hyperparameters. Specifically, the number of sliding windows T is selected from $\{2, 3, 4, 5, 6, 7, 8\}$, γ is selected from $\{1e-7, 1e-6, 1e-5, 1e-4, 1e-3\}$, α in the range of $\{0.2, 0.4, 0.5, 0.6, 0.8\}$, β and λ are varied within the range of $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1\}$. The order of neighboring nodes K is 2, and we apply one layer of HO-STGCN to extract the spatio-temporal topological features from TEGs. Next, the extracted features are flattened into a one-dimensional vector, and then the vector dimensions are mapped to 32 using a fully connected layer, batch normalization layer, and dropout ($p = 0.4$) layer in turn. Finally, we use a fully connected layer with dimensions (32, 2) and softmax to predict the class label for each subject.

We conduct a total of eight binary classification tasks on two datasets. On the ADNI dataset, the four binary classification tasks are HC vs. ILL (SMC&EMCI), HC vs. SMC, HC vs. EMCI and SMC vs. EMCI. On the PD dataset, the four binary classification tasks are: HC vs. ILL (TDPD&PIGD-PD), HC vs. TDPD, HC vs. PIGD-PD and TDPD vs. PIGD-PD. We utilize ten-fold cross-validation to measure the diagnostic performance of all methods. The performance is evaluated by accuracy (ACC), precision (PRE), recall (REC) and the area under the receiver operation characteristic curve (AUC).

D. Classification Performance

The classification results of all methods on the ADNI and PD datasets are presented in Table I and Table II, respectively. From these tables, we can draw the following two conclusions. Firstly, spatio-temporal models such as FE-STGNN, ST-GCN, ACIFBN and GHP-OT can achieve superior classification performance compared to other methods, which proves that extracting the spatio-temporal features of DFBN is important for brain disease diagnosis. Secondly, the proposed method

TABLE I
CLASSIFICATION PERFORMANCE (MEAN/STD) OF DIFFERENT METHODS ON THE ADNI DATASET (%)

Method	HC vs. ILL				HC vs. SMC			
	ACC	PRE	REC	AUC	ACC	PRE	REC	AUC
GCN	79.58/09.73	84.86/08.39	81.87/11.77	72.61/14.96	85.50/06.81	90.50/08.69	79.14/13.91	84.01/05.85
FBNetGen	81.52/07.63	88.59/08.35	81.35/16.15	81.69/09.36	84.88/04.01	94.58/08.75	69.52/09.60	81.79/09.97
CGRL	83.43/09.03	87.84/12.57	81.29/23.73	83.58/11.99	85.62/10.98	88.86/09.31	85.58/16.73	86.42/11.55
MDRL	84.33/06.18	88.28/11.81	88.60/11.97	85.98/08.88	86.71/08.80	88.53/13.84	82.20/14.68	81.97/14.16
DHOFC	73.08/09.70	77.69/10.73	78.94/12.43	78.44/12.31	73.42/11.49	73.89/13.32	66.64/20.24	78.91/15.76
Ho-D-FCNs	72.53/11.40	80.55/12.36	74.44/15.13	82.08/11.90	76.54/10.97	84.77/11.68	65.77/16.26	78.40/16.46
FE-STGNN	85.50/07.31	86.12/06.55	88.62/06.83	83.49/08.15	83.90/11.07	91.13/12.25	76.37/13.27	90.60/10.30
ST-GCN	84.21/07.00	88.94/11.03	86.18/10.76	84.20/08.78	88.76/04.78	96.21/05.83	81.11/13.96	89.35/09.11
ACIFBN	86.21/07.54	90.70/09.16	88.27/11.13	88.00/07.56	86.76/07.21	89.64/16.10	79.29/14.72	85.45/12.32
GHP-OT	87.00/09.53	87.57/12.03	88.16/11.27	86.83/09.63	87.52/07.19	90.95/12.32	84.70/11.84	87.24/07.05
MSSTAN	84.14/07.03	91.02/06.87	83.17/15.27	87.05/06.20	88.19/09.01	91.20/11.68	85.69/12.62	88.33/08.90
STIGR	82.81/04.33	87.95/07.79	85.13/11.30	84.46/09.99	88.71/05.82	92.35/09.62	85.30/11.55	92.02/04.16
Ours	87.24/05.36	88.71/10.53	91.71/08.50	91.81/03.68	91.00/06.90	94.31/07.49	88.22/15.21	94.24/06.69
Method	HC vs. EMCI				SMC vs. EMCI			
	ACC	PRE	REC	AUC	ACC	PRE	REC	AUC
GCN	75.38/10.20	81.25/17.08	80.24/16.71	67.35/14.49	75.05/05.28	75.68/12.83	75.32/11.37	67.92/12.89
FBNetGen	79.29/09.29	78.68/29.04	63.81/23.86	65.94/21.14	74.55/06.64	77.36/14.16	80.09/18.98	75.88/09.23
CGRL	84.86/07.88	90.96/10.72	76.00/22.85	83.46/09.40	81.54/09.85	79.04/15.80	85.42/16.46	76.48/11.70
MDRL	84.84/07.70	85.23/29.40	67.88/26.67	79.74/19.68	79.23/07.73	76.40/10.60	85.64/15.39	72.76/12.73
DHOFC	75.82/13.56	77.33/31.97	65.64/29.90	80.89/17.00	73.85/03.77	79.73/11.64	70.57/17.92	69.53/17.60
Ho-D-FCNs	72.86/11.10	77.17/29.78	51.19/26.30	75.24/11.70	53.33/09.49	56.53/11.54	58.91/13.22	53.29/15.71
FE-STGNN	78.46/07.54	77.19/12.00	84.79/16.71	70.02/09.82	76.67/12.80	73.94/27.15	68.21/32.30	60.61/17.28
ST-GCN	84.50/06.87	88.36/11.26	87.72/12.74	86.42/09.79	81.54/07.84	83.70/10.43	81.52/20.09	72.64/12.57
ACIFBN	85.00/06.24	83.87/13.06	87.38/13.38	73.87/17.76	80.77/12.99	79.86/16.25	88.89/10.83	80.81/10.52
GHP-OT	84.95/07.38	88.57/10.87	76.60/21.61	85.65/10.47	80.00/06.15	79.35/12.61	86.85/10.66	74.31/12.30
MSSTAN	85.82/06.88	84.95/10.58	83.75/22.70	82.93/10.79	76.92/07.69	84.89/12.51	76.71/26.92	68.96/15.38
STIGR	79.84/05.24	78.59/29.07	61.19/27.52	69.18/17.35	76.15/08.74	78.15/12.29	80.86/11.88	76.04/14.22
Ours	86.59/11.13	85.76/19.04	90.38/12.67	90.05/14.70	86.81/05.68	87.50/09.43	89.40/14.74	87.08/07.98

outperforms all others in all classification tasks. Specifically, for the ADNI dataset, the accuracy of our method on the four tasks is improved by 0.24%, 2.24%, 1.59% and 5.27%, respectively. For the PD dataset, the accuracy of our method on the four tasks is improved by 1.25%, 0.78%, 3.41% and 3.73%, respectively. In addition, we plot the ROC curves of the different methods in Figure 4. It can be seen from the figure that the ROC curve of the proposed method is located in the upper left corner of all methods, which verifies that this method has the highest AUC value. It is worth noting that the proposed method also achieves almost optimal performance on other evaluation metrics, which provides comprehensive evidence for its effectiveness.

Our method's advantage lies in its ability to capture the topological evolution process simultaneously at the node and edge levels, and reveal the internal evolutionary structure based on PRI, which helps the HO-STGCN efficiently extract the dynamic topological features in TEGs. Static analysis methods like GCN, FBNetGen and CGRL only focus on topological structural features and ignore temporal information in the brain. Dynamic analysis methods such as DHOFC, ACIFBN and GHP-OT mainly capture the dynamic changes of brain region signals, ignoring the topological edges also change over time. In addition, these methods fail to effectively mitigate the redundant connections of brain networks,

potentially introducing interference when aggregating dynamic features. In contrast, our method not only integrates the spatio-temporal evolution laws of the brain at the node and edge levels, but also effectively reveals the internal structure of the brain.

E. Ablation Experiment

The proposed framework contains three modules: TEG, PRI and HO-STGCN. To assess the contribution of each module, we conduct multiple ablation experiments on the ADNI and PD datasets, and the experimental results are shown in Table III. The simplified models we employed for these experiments include: (1) baseline: Multi-layer perceptron is utilized to extract features from DFBNs. (2) w/o TEG&PRI: This model uses only HO-STGCN to extract features from DFBNs. (3) w/o TEG&HG: This model employs PRI to alleviate redundant information in DFBN and uses GCN to extract features. (4) w/o PRI&HG: This model uses GCN to extract features from TEG without eliminating redundant information. (5) w/o TEG: This model replaces the TEG with the original DFBNs. (6) w/o PRI: This model removes the PRI, using TEG without eliminating redundant information for feature extraction. (7) w/o HO-STGCN: This model turns HO-STGCN into GCN.

TABLE II
CLASSIFICATION PERFORMANCE (MEAN/STD) OF DIFFERENT METHODS ON THE PD DATASET (%)

Method	HC vs. ILL				HC vs. TDPD			
	ACC	PRE	REC	AUC	ACC	PRE	REC	AUC
GCN	75.29/10.09	82.08/10.20	80.38/09.51	68.53/16.53	78.44/09.79	76.24/29.19	63.57/25.92	77.58/12.22
FBNetGen	66.62/08.10	79.97/11.42	68.13/15.72	49.01/21.66	74.56/08.13	65.50/36.07	51.28/34.88	65.65/23.42
CGRL	76.47/08.44	80.45/09.28	86.32/22.47	61.13/21.34	77.44/10.54	76.22/30.71	62.89/29.84	58.75/31.96
MDRL	75.88/09.53	79.01/10.75	87.84/19.52	69.03/16.50	76.75/10.61	78.67/16.88	65.81/21.33	60.63/20.99
DHOFc	67.94/08.32	73.21/11.36	86.81/28.09	53.65/14.90	70.44/12.12	68.54/28.87	61.38/29.40	61.68/23.05
Ho-D-FCNs	67.50/06.12	77.74/11.35	72.17/12.19	67.01/11.01	66.45/10.70	63.92/36.87	50.33/41.91	68.32/17.84
FE-STGNN	79.63/04.82	82.47/09.27	87.89/07.13	73.05/07.48	81.78/09.63	68.00/35.00	64.67/34.96	75.58/18.68
ST-GCN	79.67/08.53	84.35/10.74	86.08/20.38	80.81/13.60	82.67/09.01	83.83/13.72	82.33/16.74	80.18/13.04
ACIFBN	79.60/05.64	85.40/09.96	83.46/13.30	71.21/13.65	82.89/10.93	79.71/16.76	81.33/21.66	79.80/18.10
GHP-OT	82.10/08.59	84.82/09.16	86.02/23.01	78.84/11.71	81.56/10.89	83.98/13.77	79.33/20.75	81.59/13.40
MSSTAN	81.58/09.37	85.17/09.32	84.96/18.85	74.58/15.26	80.67/12.54	85.00/18.44	70.50/23.62	79.25/14.74
STIGR	77.79/05.55	81.36/12.49	88.94/14.14	70.01/15.02	78.67/13.67	65.87/32.00	80.00/31.44	81.64/18.57
Ours	83.35/07.83	86.50/09.10	90.52/11.18	81.45/14.12	83.67/08.06	85.81/15.21	83.14/19.43	82.63/14.07
Method	HC vs. PIGD-PD				TDPD vs. PIGD-PD			
	ACC	PRE	REC	AUC	ACC	PRE	REC	AUC
GCN	71.50/13.30	69.84/20.04	82.32/11.24	68.58/16.52	75.72/09.07	81.06/16.00	72.09/17.05	69.97/19.59
FBNetGen	74.55/07.51	76.67/13.84	84.52/13.93	51.82/25.05	72.00/12.49	80.02/17.00	71.29/20.94	66.14/19.14
CGRL	76.36/07.93	81.40/12.69	77.86/16.19	69.20/08.19	74.00/09.17	75.14/13.79	84.55/14.63	54.72/13.06
MDRL	75.50/10.06	75.04/11.90	79.90/19.35	70.98/10.49	73.09/09.61	79.35/14.91	82.90/27.49	50.81/22.08
DHOFc	70.53/09.67	74.26/11.09	69.39/15.64	66.79/11.29	65.00/10.25	74.03/13.62	67.05/17.74	55.44/13.48
Ho-D-FCNs	67.88/09.45	80.14/15.53	58.58/18.89	61.21/14.72	62.45/14.51	61.74/14.97	84.57/27.97	52.13/11.64
FE-STGNN	78.11/06.41	85.16/12.93	70.17/20.45	73.60/12.50	70.55/12.57	69.41/13.97	85.58/17.55	56.08/17.87
ST-GCN	79.70/10.05	89.78/10.95	69.49/30.31	77.15/13.92	74.45/14.44	66.81/27.85	82.08/31.18	67.44/18.90
ACIFBN	76.97/13.27	70.47/27.91	80.32/29.60	68.59/24.62	74.18/14.29	69.61/26.71	79.19/28.61	63.79/10.20
GHP-OT	82.20/08.69	84.73/14.21	85.86/14.92	75.50/15.85	75.82/11.14	67.02/24.63	83.33/29.81	63.05/18.53
MSSTAN	78.86/08.57	84.85/13.99	76.67/24.93	70.36/21.56	77.55/10.85	79.61/17.09	76.52/25.27	71.71/24.00
STIGR	75.53/08.61	71.55/25.88	72.19/33.11	68.60/15.84	76.21/07.42	75.89/12.64	84.50/19.26	76.57/09.93
Ours	85.61/06.52	84.36/11.88	90.32/11.01	84.91/12.44	79.55/09.14	86.59/12.30	81.49/22.49	75.78/15.20

TABLE III
ABLATION EXPERIMENTAL PERFORMANCE UNDER DIFFERENT TASKS ON ADNI AND PD DATASETS (%)

Dataset	Method	HC vs. ILL				HC vs. SMC				HC vs. EMCI				SMC vs. EMCI			
		ACC	PRE	REC	AUC	ACC	PRE	REC	AUC	ACC	PRE	REC	AUC	ACC	PRE	REC	AUC
ADNI	baseline	82.50	84.79	88.03	81.08	84.62	93.18	73.38	83.79	81.92	82.92	63.01	79.37	72.31	70.06	85.65	67.71
	w/o TEG&PRI	84.74	85.33	89.40	86.46	87.14	87.81	83.85	91.95	83.35	90.07	72.51	82.18	73.85	75.86	80.27	73.72
	w/o TEG&HG	84.83	84.78	93.61	87.64	88.00	93.68	81.35	90.81	82.19	85.59	75.07	79.92	73.08	66.00	79.56	58.67
	w/o PRI&HG	84.71	84.96	92.30	83.97	86.76	86.50	82.65	89.25	82.75	83.57	72.02	83.28	75.38	78.04	77.67	58.75
	w/o TEG	85.00	85.84	92.25	84.16	88.10	87.32	86.77	91.78	86.48	87.46	80.82	89.88	76.15	67.15	78.68	71.34
	w/o PRI	87.00	88.63	89.85	87.08	89.48	90.92	85.88	91.19	85.05	88.02	76.29	87.36	77.69	77.18	79.79	66.24
	w/o HG	85.50	86.88	88.03	87.56	89.52	91.70	87.17	89.53	83.57	85.61	79.10	85.24	76.92	70.11	73.67	65.42
	ours	87.24	88.71	91.71	91.81	91.00	94.31	88.22	94.24	86.59	85.76	90.38	90.05	86.81	87.50	89.40	87.08
PD	baseline	73.82	77.55	72.67	69.53	73.13	82.51	74.42	77.80	72.83	67.33	73.55	56.72	73.91	61.70	75.48	62.61
	w/o TEG&PRI	78.35	86.27	81.95	72.06	76.44	65.57	59.50	68.85	72.88	67.28	75.07	64.23	73.09	73.99	85.32	60.89
	w/o TEG&HG	78.27	83.12	87.91	71.36	76.33	74.28	60.42	70.42	74.55	78.31	66.99	72.26	73.27	71.88	89.14	60.65
	w/o PRI&HG	77.17	85.54	82.64	77.59	79.78	74.57	71.33	74.10	74.09	68.69	72.57	55.36	72.36	67.11	82.38	61.71
	w/o TEG	78.42	81.10	86.01	72.30	79.56	82.00	73.00	75.15	76.21	84.06	75.67	71.29	74.82	77.14	68.57	59.02
	w/o PRI	80.18	82.02	89.74	73.04	81.44	79.84	80.00	77.38	77.64	85.23	79.51	76.04	72.27	78.75	79.24	50.00
	w/o HG	81.36	84.31	87.19	76.01	82.67	87.33	80.33	79.33	77.95	79.73	85.91	69.89	75.00	75.22	70.00	51.83
	ours	83.35	86.50	90.52	81.45	83.67	85.81	83.14	82.63	85.61	84.36	90.32	84.91	79.55	86.59	81.49	75.78

w/o means without, and HG means HO-STGCN.

From Table III, we can draw the following conclusions. First, compared with the baseline, each module can significantly improve diagnostic performance. Second, retaining two modules simultaneously yields higher diagnostic accuracy

than retaining only one module, suggesting that the proposed modules are mutually beneficial. For example, for the ADNI dataset, the w/o TEG method achieved 0.26%, 0.96%, 3.13%, and 2.30% higher accuracy than the w/o TEG&PRI method

TABLE IV

THE CLASSIFICATION ACCURACY OF ABLATION STUDY ABOUT WD AND GWD ON THE ADNI AND PD DATASET (%)

Method	HC vs. ILL	HC vs. SMC	HC vs. EMCI	SMC vs. EMCI
w/o WD	85.50	88.86	85.66	77.62
w/o GWD	86.00	91.62	86.54	78.46
ours	87.24	91.00	86.59	86.81
Method	HC vs. ILL	HC vs. TPD	HC vs. PIPD	TPD vs. PIPD
w/o WD	79.01	81.00	84.05	75.91
w/o GWD	79.63	82.78	81.52	76.18
ours	83.35	83.67	85.61	79.55

w/o means without, TPD means TDPD, PIPD means PIGD-PD.

on the four tasks, respectively. The w/o TEG method also achieved 0.17%, 0.10%, 4.29%, and 3.07% higher accuracy than the w/o TEG&HG method on the four tasks, respectively. Moreover, the recognition performance is optimal when all modules are integrated. This is because TEG integrates spatio-temporal evolution information at both the node and edge levels. PRI further alleviates the redundant features in TEG, thereby revealing the dynamic structures closely related to brain diseases. Finally, HO-STGCN aggregates the high-order spatio-temporal correlations of neighboring nodes at different distances in TEG. Therefore, the three proposed modules can effectively represent and extract the dynamic topological information of brain networks, thereby significantly improving classification accuracy.

Furthermore, we explore the impact of node-level evolution and edge-level evolution on brain disease diagnosis. Specifically, we maintain the same experimental setup and only consider whether WD and GWD are retained, and the classification results are demonstrated in Table IV. Table IV shows that either removing WD or GWD will reduce the classification accuracy, with the combination of both yielding optimal performance. This phenomenon suggests that the proposed WD and GWD can comprehensively capture the evolution law of brain regions and brain connections, thereby improving the diagnostic effect of brain diseases.

V. DISCUSSION

A. Temporal-Spatial Variability of TEGs

To investigate the impact of brain diseases on the evolution characteristics of brain network, we take the variance of connection strength of each brain region in TEG under different windows as the spatio-temporal variability, and verify whether the change of spatio-temporal properties by brain disease is significant by *t*-test. As shown in Figure 5, brain diseases make the topological structure of brain networks more unstable. Specifically, the spatio-temporal variability of patients with cognitive impairment and Parkinson's disease is significantly higher than that of HC. For the ADNI dataset, this phenomenon may result from the interruption or abnormal enhancement of brain network connectivity triggered by cognitive impairment, which leads to the decrease of brain coordination in spatio-temporal dimensions [44]. In addition, cognitive impairment is usually accompanied by structural degradation of brain regions and abnormal neurotransmitter function, which interferes with normal communication

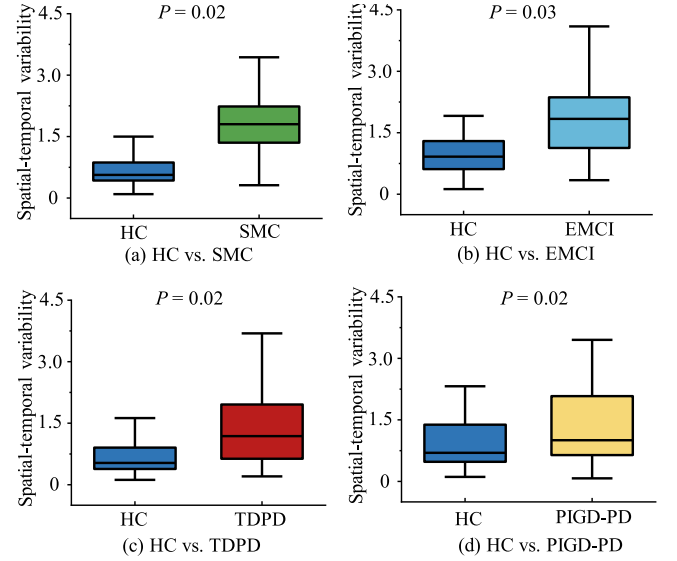


Fig. 5. Analysis of spatio-temporal variability under different diagnostic tasks.

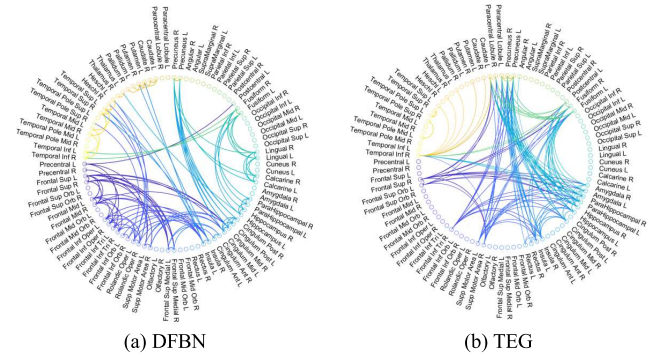


Fig. 6. The 120 most important brain connections on the HC vs. SMC task.

between brain regions and makes brain region activity more disordered [45]. For the PD dataset, this phenomenon occurs because the neural circuits of PD patients are disrupted, and this pathological process affects the reorganization and dysfunction of the global network through long-distance connections, resulting in a decrease in the stability of the brain network [46]. Understanding this characteristic is beneficial to reveal the neural mechanism of diseases, and also provides new directions for its diagnosis and intervention.

B. Discriminative Brain Connections

To further validate the effectiveness of the proposed method in detecting biomarkers, we visualize the top 120 brain connections with the largest weights from both the original DFBN and the TEG refined by PRI. The key brain connections on the NC vs. SMC task and NC vs. EMCI task are shown in Figure 6 and Figure 7, respectively. These visualizations allow us to draw two main conclusions. Firstly, DFBN focuses more on connections between physically adjacent regions, whereas the proposed TEG emphasizes long-range connections between hub nodes. DFBN typically calculates pairwise correlations between brain regions, which amplifies

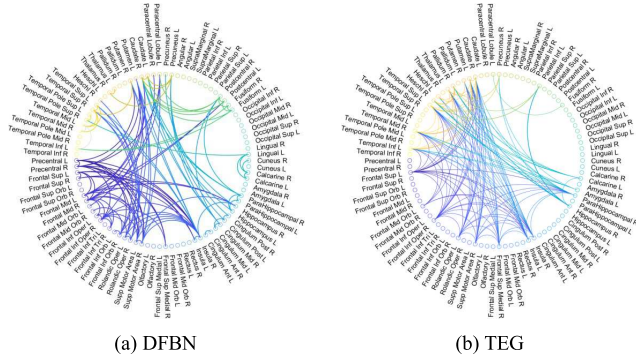


Fig. 7. The 120 most important brain connections on the HC vs. EMCI task.

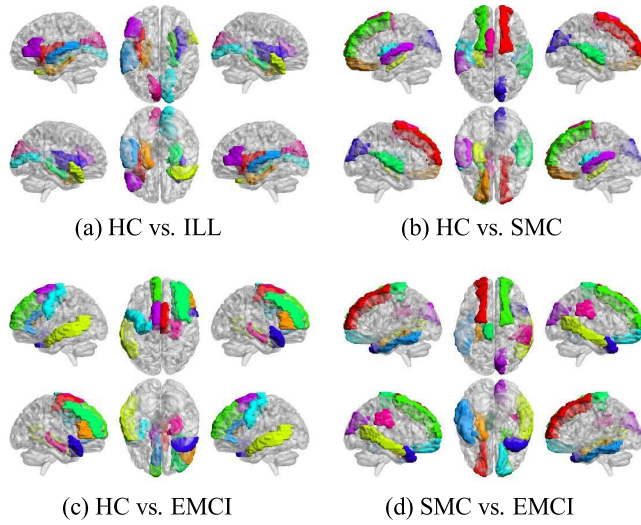


Fig. 8. Distribution locations of the 10 most discriminative brain regions under different tasks on the ADNI dataset.

the functional connections of neighboring areas while neglecting long-range interactions. In contrast, the proposed TEG characterizes the evolution of brain networks at both the node and edge levels through the propagation of hubness, capturing cross-spatiotemporal interactions between hub nodes. This is beneficial for representing the neural mechanisms underlying cross-regional collaboration in the brain. Secondly, the identified key brain connections are closely related to the occurrence of cognitive impairment. Specifically, for NC vs. SMC, key brain connections are concentrated in the Frontal.Sup.R, Supplementary Motor Area, Hippocampus and other regions. SMC can cause the decline of motor control and memory ability, so the brain promotes the normal operation of function by enhancing the corresponding information transmission mechanism [24], [47]. For NC vs. EMCI, key brain connections are focused on Frontal.Sup.L, Frontal.Inf.Oper.R, and Insula. The observed phenomenon could be due to the gradual cognitive decline in EMCI patients, leading to abnormal function in the regions responsible for cognitive processes [48].

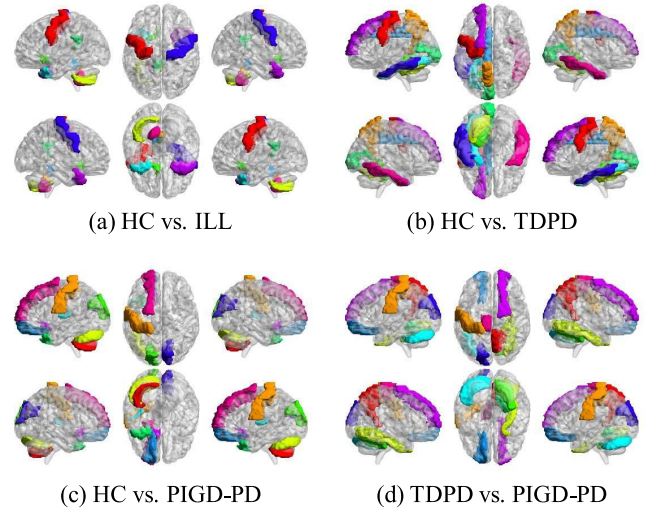


Fig. 9. Distribution locations of the 10 most discriminative brain regions under different tasks on the PD dataset.

C. Discriminative Brain Regions

For each diagnosis task, we calculate the variance of connection strength across all brain regions in TEGs, and sum it by row to obtain the spatio-temporal variation rate of each brain region. Then, we visualize the 10 brain regions with the highest spatio-temporal variability under different tasks in Figure 8. For NC vs. ILL, important brain regions include Hippocampus, Superior temporal gyrus and Insula, which are related to memory and cognitive regulation. This emphasis may stem from the disruption of normal cognitive functions due to cognitive disorders [24]. For NC vs. SMC, important regions are concentrated in the Frontal.Sup.R, Hippocampus, and Temporal.Sup.R. The presence of significant memory defects in SMC likely leads to functional abnormalities in these areas [49]. For NC vs. EMCI, unstable brain areas mainly include Supplementary motor area, Inferior frontal gyrus and Middle temporal gyrus. This may be because EMCI causes abnormalities in brain processing of complex cognitive, motor, and emotional tasks [22]. For SMC vs. EMCI, important brain regions are concentrated in the frontal and temporal lobes, such as Superior frontal gyrus, Middle temporal gyrus and Inferior temporal gyrus. The frontal and temporal lobes control cognition and decision making, respectively. The abnormal interactions in these regions may be an important indicator of the progression from SMC to EMCI [3], [50].

From Figure 9, we can observe that the cerebellum plays a significant role in the diagnosis of Parkinson's disease. Specifically, for HC vs. ILL, the most important brain regions include the Precentral Gyrus, Vermis.10, and Cerebellum.9.L. This is because Parkinson's disease leads to cognitive decline and motor dysfunction, causing abnormalities in brain regions responsible for these functions [51]. For HC vs. TDPD, the most important brain regions include the Cingulum.Mid.L, Inferior Temporal Gyrus, and Cerebellum.6.L. Patients with TDPD often experience confusion and motor impairments, which may be due to abnormalities in these brain regions responsible for visual processing and motor control [52],

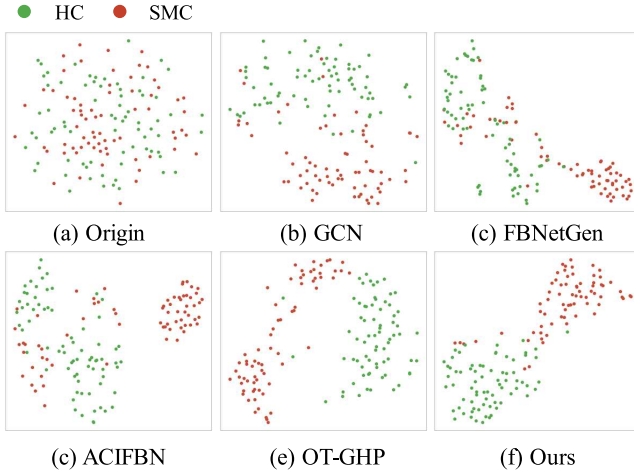


Fig. 10. T-SNE visualization for different methods on the HC vs. MCI task.

[53]. For HC vs. PIGD-PD, the significant brain regions include Olfactory.L, Cerebellum.7b.L, and Cerebellum.Crus1.L. Patients with PIGD-PD exhibit olfactory dysfunction and gait disorders [54], [55]. Therefore, these brain regions responsible for olfactory processing and gait control can serve as key biomarkers for identifying PIGD-PD. For TDPD vs. PIGD-PD, the key brain regions are Precuneus.R, Cerebellum.Crus1.L, and Cerebellum.6.R. Patients with TDPD primarily exhibit prominent tremor symptoms, while PIGD-PD is mainly characterized by postural instability and gait disorders [52], [54]. It is worth noting that these brain regions are closely related to motor control, making them discriminative brain regions for distinguishing between the two diseases.

D. T-SNE Visualization

To visually demonstrate the performance of different methods, we use t-SNE to visualize their learned features. As shown in Figure 10, the original feature distribution appears highly disordered. GCN, FBNetGen, and ACIFBN form two clusters with confusion. Although OT-GHP can distinguish between the two types of subjects to some extent, the learned feature distribution is loose and fails to establish clear inter-class boundaries. In contrast, the proposed method not only effectively aggregates intra-class features but also maintains distinct inter-class separation. This is because GCN and FBNetGen ignore the dynamic properties of the brain, while ACIFBN and OT-GHP focus solely on capturing the temporal evolution at the node level. It is worth noting that the proposed method can not only collaboratively model the high-order topological evolution at both the node and edge levels of the brain, but also alleviates redundant information unrelated to disease in TEGs. Therefore, the features learned by the proposed method can better distinguish subjects across different types.

E. Parameter Sensitivity Analysis

To validate the robustness of the proposed method, we conduct sensitivity analyses on the hyperparameters γ , α , β , and λ . In Figure 11 (a), the highest accuracy is achieved

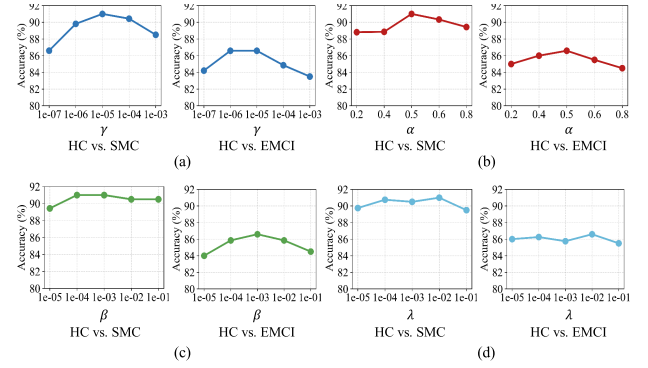


Fig. 11. The impact of hyperparameters on diagnostic performance. (a) The impact of γ on accuracy. (b) The impact of α on accuracy. (c) The impact of β on accuracy. (d) The impact of λ on accuracy.

TABLE V
COMPUTATIONAL EFFICIENCY OF THE PROPOSED METHOD AND OTHER SPATIO-TEMPORAL FEATURE EXTRACTION METHODS

Method	Param (M)	FLOPs (M)	Inference time (s)
FE-STGNN	2.8017	2.8593	0.0078
ST-GCN	2.7978	5.8566	0.0012
ACIFBN	3.9087	20.7483	0.0123
GHP-OT	4.1417	6.9713	0.0050
MSSTAN	4.6665	14.9545	0.0023
STIGR	10.6194	30.2721	0.0219
Ours	2.7721	4.6380	0.7553

when $\gamma = 1e-5$. This is because a very small γ retains more noisy edges, while a very large γ reduces topological information. When $\gamma = 1e-5$, it can effectively reduce noise while preserving the key topological information. In Figure 11 (b), the optimal performance is achieved when $\alpha = 0.5$. This is because α controls the relative importance between node transport cost and edge transport cost. When α is too large, the model overly focuses on edge evolution; when α is too small, the model mainly focuses on node evolution. A moderate $\alpha = 0.6$ achieves a good balance between nodes and edges, thereby improving overall performance. In Figure 11 (c), the best result is achieved when $\beta = 1e-3$. This is because β measures the relative importance of structure retention and redundancy reduction. When β is set to $1e-3$, it can preserve discriminative information while mitigating structural redundancy, thereby better reflecting the intrinsic spatio-temporal evolution of the brain. Additionally, in Figure 11 (d), we can see that the accuracy remains consistently high as λ changes, indicating that the proposed method is not sensitive to λ . Overall, the accuracy of the proposed method does not fluctuate significantly with changes in hyperparameters, demonstrating its robustness.

F. Computational Efficiency Analysis

To investigate the computational efficiency of the model, we compare the model parameters (Param), floating-point operations (FLOPs), and the average inference time required per subject of the proposed method and some spatio-temporal methods. As shown in Table V, the model parameters of this

method amount to 2.7721 million, which is the fewest among all spatio-temporal methods. Additionally, our method involves only 4.638 million FLOPs, achieving performance second only to FE-STGNN. Although the inference time required by our method is 0.7553 seconds, which is slightly longer than other spatio-temporal models, it achieves the highest classification accuracy. Therefore, our method can achieve superior diagnostic outcomes using relatively fewer computational resources.

VI. CONCLUSION

In this paper, we propose a topological evolution graph learning model to capture the complex spatio-temporal evolution characteristics of dynamic brain networks. Firstly, the hubness distributions of adjacent brain networks are taken as the source domain and target domain in turn, and WD and GWD are used to adaptively learn the topology evolution process at the node level and the edge level, respectively. Then, to further reveal the dynamic evolution information related to brain diseases, we adopt PRI to guide TEG to learn a structure containing the most relevant but least redundant information, aiming to preserve the valuable spatio-temporal features in the brain network. Finally, we design a high-order spatio-temporal model with multi-hop GCN to collaboratively extract long-range spatial correlation and dynamic temporal features from TEGs. Extensive experiments show that the proposed method is superior to other brain network analysis methods, and can effectively reveal the information propagation mechanism between cross-window brain regions.

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